

fundamental necessity in data science. CSV is a simple file format used to store tabular data, such as a spreadsheet or database. The standard CSV format is defined by row and column data. Each row is terminated by a new line to begin the next row. Also, within the row, each column is separated by a comma.

For working with CSV files in Python, import an inbuilt module called `csv` as follows:

```
import csv
# reading csv file
with open(':/Users/Neethu/data.csv','r') as x:

# creating a csv reader object
data = csv.reader(x)
# extracting each data row one by one
for row in data:
    print(row)
# print the total number of rows
print("Total no. of rows: %d" %(data.line_num))
```

The output is given below:

```
['id', 'name', 'salary', 'start_date', 'dept']
['1', 'John', '34000', '05-08-17', 'CSE']
['2', 'Manu', '25600', '23-09-18', 'ECE']
['3', 'Isha', '32500', '05-12-18', 'IT']
['4', 'Ryan', '35678', '01-05-19', 'CSE']
['5', 'Neha', '32100', '27-03-17', 'ME']
['6', 'Pooja', '31000', '03-05-19', 'IT']
['7', 'Sam', '26000', '30-07-17', 'ECE']
['8', 'Sanju', '35908', '27-06-18', 'CSE']
Total no. of rows: 9
```

The Pandas open source library also provides features using which we can read the CSV file in full as well as in parts for only a selected group of columns and rows. Reading the CSV into a Pandas DataFrame is very quick and easy:

```
import pandas as pd
x = pd.read_csv('C:/Users/Neethu/data.csv')
print(x)
```

The `read_csv` function of the Pandas library reads the files from the OS by using the proper path to the file. An additional column starting with zero as an index is created by the function.

**Reading specific rows and columns:** The code `print(x[0:5]['salary'])` will slice the result for the first rows. The `read_csv` function can also be used to read some specific columns along with the multi-axes indexing method called `.loc()`.

|   | id | name  | salary | start_date | dept |
|---|----|-------|--------|------------|------|
| 0 | 1  | John  | 34000  | 05-08-17   | CSE  |
| 1 | 2  | Manu  | 25600  | 23-09-18   | ECE  |
| 2 | 3  | Isha  | 32500  | 05-12-18   | IT   |
| 3 | 4  | Ryan  | 35678  | 01-05-19   | CSE  |
| 4 | 5  | Neha  | 32100  | 27-03-17   | ME   |
| 5 | 6  | Pooja | 31000  | 03-05-19   | IT   |
| 6 | 7  | Sam   | 26000  | 30-07-17   | ECE  |
| 7 | 8  | Sanju | 35908  | 27-06-18   | CSE  |

Figure 1: Reading CSV file using `read_csv()` of Pandas

```
print(x.loc[2:5,['salary','name']])
salary name
2 32500 Isha
3 35678 Ryan
4 32100 Neha
5 31000 Pooja
```

## Python: Processing JSON data

A JSON file stores data as text in a human-readable format. JSON stands for JavaScript Object Notation. Pandas can read JSON files using the `read_json` function. Python is supported with the JSON library. We can convert lists and dictionaries to JSON, and strings to lists and dictionaries. JSON data looks much like a dictionary would in Python, with the keys and values stored.

```
import json
data = '''{"apple": "red", "banana": "yellow", "mango": "green"}'''
# load the JSON into 'a' variable.
a = json.loads(data)
print(a)
#Write the Data into a JSON File
with open('abc.json', 'w') as f:
    json.dump(a, f)
```

The output is shown below:

```
{'apple': 'red', 'banana': 'yellow', 'mango': 'green'}
```

```
Using pandas, read a json file using the function pd.read_
json()
import pandas as pd
newdata = pd.read_json('C:/Users/Neethu/inputdata.json')
print(newdata)
```

## Python: Processing XLSX data

The Pandas library provides features using which we can read the Excel file in full as well as in parts, for only a selected group of data. We can also read an Excel file with